# Mass - Spring - Damper System Simulation

```
clc;
clear;
close all;
% System Parameters
m = 1;       % Mass (kg)
c = 2;       % Damping coefficient (Ns/m)
k = 20;      % Spring stiffness (N/m)
% Time range
tspan = 0:0.01:10;
% Initial conditions
x0 = [0; 0];
% Differential equation:
% m*x'' + c*x' + k*x = 1
% Step Response
[t, x] = ode45(@(t,x) [x(2);
    (1-c*x(2)-k*x(1))/m], tspan, x0);
% Displacement response
y = x(:,1);
figure;
plot(t,y,'b','LineWidth',1.5);
grid on;
xlabel('Time (seconds)');
ylabel('Displacement (m)');
title('Step Response of Mass-Spring-Damper System');
```

```
% Performance Parameters
yss = 1/k;
peak_value = max(y);
overshoot = max(0,(peak_value-yss)/yss*100);
peak_index = find(y==peak_value,1);
peak_time = t(peak_index);
idx10 = find(y >= 0.1*yss,1);
idx90 = find(y >= 0.9*yss,1);
if ~isempty(idx10) && ~isempty(idx90)
    rise_time = t(idx90)-t(idx10);
else
    rise_time = NaN;
end
idx_settle = find(abs(y-yss)>0.02*abs(yss),1,'last');
if isempty(idx_settle)
    settling_time = 0;
elseif idx_settle < length(t)
    settling_time = t(idx_settle+1);
else
    settling_time = NaN;
end
disp('Performance Parameters:');
fprintf('Steady State Value: %.4f m\n',yss);
fprintf('Peak Value: %.4f m\n',peak_value);
fprintf('Peak Time: %.4f seconds\n',peak_time);
```

```
fprintf('Overshoot: %.2f %%\n',overshoot);
fprintf('Rise Time: %.4f seconds\n',rise_time);
fprintf('Settling Time: %.4f seconds\n',settling_time);
% Effect of Different Damping Values
figure;
hold on;
c_values = [0.5 2 8];
for c = c_values
    [t,x] = ode45(@(t,x) [x(2);
        (1-c*x(2)-k*x(1))/m],tspan,x0);
    plot(t,x(:,1),'LineWidth',1.5);
end
legend('c=0.5','c=2','c=8');
xlabel('Time (seconds)');
ylabel('Displacement (m)');
title('Effect of Damping Coefficient');
grid on;
hold off;
```

**Simulink**

